

INTRODUCTION TO STATISTICAL METHODS

Analysis of Variance (ANOVA)

Comparing several group means with the F-ratio: between-group vs within-group variance.

Companion reader · Read beside the slides

How to use this companion. Colour boxes guide you: **blue** intuition, **amber** everyday picture, **green** worked example, **red** watch out, **purple** key takeaway. Read it next to the slides, in order.

1 Why not many t-tests?

A t-test compares *two* means. To compare *three or more* groups you might run a t-test on every pair — but each test carries a Type I risk, and the chance of *some* false alarm balloons. **Analysis of Variance** (ANOVA) tests all the means *at once* with a single test, keeping the error rate controlled.

The intuition

ANOVA asks: is the spread *between* the group means large compared with the random spread *within* each group? If the groups are drawn from the same population, between-group spread should be about the same as within-group spread. If between is much larger, the groups really differ.

2 Partitioning the variance

ANOVA splits total variation into two parts and forms a ratio:

$$F = \frac{MSB}{MSW} = \frac{\text{between-group variance}}{\text{within-group variance}},$$

where $MSB = SSB/(k - 1)$ and $MSW = SSW/(N - k)$ for k groups and N total observations. **SSB** is the between-group sum of squares (group means vs the grand mean); **SSW** is the within-group sum of squares (points vs their own group mean). Under H_0 (all means equal), $F \approx 1$; a large F means reject.

Worked example: one-way ANOVA by hand

Three groups: $A = \{1, 2, 3\}$ (mean 2), $B = \{4, 5, 6\}$ (mean 5), $C = \{7, 8, 9\}$ (mean 8). Grand mean = 5. *Between*: $SSB = 3[(2-5)^2 + (5-5)^2 + (8-5)^2] = 3[9 + 0 + 9] = 54$; $df = k - 1 = 2$, so $MSB = 54/2 = 27$. *Within*: each group's SS is $(\pm 1)^2 + 0 + (\mp 1)^2 = 2$, so $SSW = 2+2+2 = 6$; $df = N - k = 6$, so $MSW = 6/6 = 1$.

$$F = \frac{27}{1} = 27.$$

That is far above 1 (and above the critical $F_{2,6} \approx 5.14$), so we **reject** equal means — the groups differ.

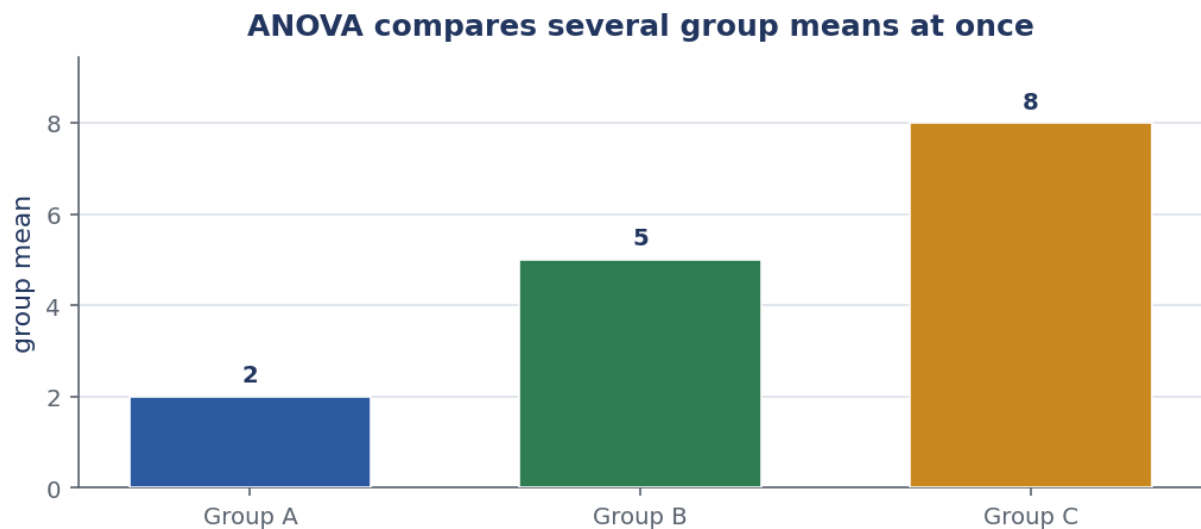


Figure 1: ANOVA asks whether the gaps *between* these group means are large relative to the scatter *within* each group.

3 Assumptions and what comes next

One-way ANOVA assumes the groups are *independent*, roughly *normal*, with similar variances (**homoscedasticity**). It tells you *that* some means differ, not *which*; **post-hoc** tests (e.g. Tukey's) then pinpoint the pairs. **Two-way ANOVA** extends this to two factors and their interaction.

✓ Key takeaway

ANOVA compares ≥ 3 group means with one test via $F = \text{MSB}/\text{MSW}$ — between-group variance over within-group variance. $\text{MSB} = \text{SSB}/(k-1)$, $\text{MSW} = \text{SSW}/(N-k)$. $F \approx 1$ supports equal means; a large F rejects (worked: $\text{SSB}=54$, $\text{SSW}=6$, $F = 27 \Rightarrow \text{reject}$). Post-hoc tests find *which* groups differ.